

Normalizing Flows for Detector-Level Jet Generation in Fully Hadronic $t\bar{t}$ Events

Jiaxun Lu

May 20, 2026

This thesis is submitted
in partial fulfillment of the requirements for
Undergraduate Departmental Honors in Physics
at the University of Washington
in Spring 2026

Approved by Research Supervisor: Prof. Gordon Watts, [DATE]

Approved by Physics 488 Instructor: Prof. Aurel Bulgac, [DATE]

© Copyright 2026

Jiaxun Lu

University of Washington

Abstract

Normalizing Flows for Detector-Level Jet Generation in Fully Hadronic $t\bar{t}$ Events

Jiaxun Lu

Modern particle physics experiments rely on detailed simulations to model high-energy collisions, but producing large realistic datasets can be computationally expensive. This thesis studies normalizing flows as likelihood-based generative models for detector-level fully hadronic $t\bar{t}$ events in ATLAS at the Large Hadron Collider. The model is trained on reconstructed jet features with the goal of generating synthetic events that reproduce both individual feature distributions and correlations among jets. Model performance is evaluated using feature histograms, correlation matrices, and physics-motivated observables. In particular, reconstructed top quark mass distributions are used to test whether generated jets preserve event-level structure relevant for analysis. The results show that normalizing flows can reproduce kinematic features, but discrepancies in correlations can significantly affect reconstructed W boson and top quark mass distributions. This work highlights both the potential of flow-based generative models for fast jet simulation and the importance of validating such models with downstream physics observables.

Acknowledgements

I would like to express my sincere gratitude to Professor Gordon Watts for their guidance, encouragement, and valuable feedback throughout this project. I would also like to thank my family and friends for their support during my undergraduate studies.

Contents

1	Introduction	4
1.1	Previous Works	5
1.2	Thesis Overview	6
2	Background	7
2.1	Detector Coordinate System	7
2.2	Jets	7
2.3	Top Quark Pair Production	8
2.4	Top Quark Mass Reconstruction	9
2.5	Normalizing Flows	11
3	Methodology	13
3.1	Dataset Overview	13
3.2	Data Preprocessing	14
3.3	Model Setup and Training	15
4	Results	17
4.1	Kolmogorov-Smirnov Statistic	17
4.2	Results and Discussions	17
5	Conclusion	21

1. Introduction

Elementary particles are the fundamental building blocks of the universe. The physics theory behind them is called the Standard Model of particle physics, which describes the elementary particles that make up matter and the fundamental interactions between them, except for gravity. In this framework, matter is built from quarks and leptons, and forces are mediated by particles called bosons, such as photons, gluons, and the weak bosons (W and Z bosons). To study these particles experimentally, physicists need to create conditions in which a large amount of energy is concentrated into a very small region. Einstein's mass-energy equivalence relation, $E = mc^2$, explains why this is possible: energy can be converted into mass, so sufficiently energetic collisions can produce massive particles that were not present before the collision. Proton-proton collisions, such as those in ATLAS, the largest general-purpose particle detector experiment at the Large Hadron Collider (LHC), are therefore a powerful tool for particle physics because protons can be accelerated to extremely high velocities by electric fields before being collided. The energy released in these collisions can produce short-lived particles whose properties are then inferred from their decay products, allowing physicists to test the Standard Model and search for possible signs of new physics.

To interpret signals of the decay products from the proton-proton collisions in ATLAS, huge amounts of simulated events are needed to compare theoretical models with experimental data, optimize algorithms, and test detector performance. Traditionally, the Monte Carlo simulation

production chain in ATLAS consists of four stages: event generation, detector simulation, digitization, and reconstruction (Table 1.1). While highly accurate and physically motivated, this production chain is also computationally expensive, especially at the scale required for modern LHC analyses. Thus, there is a strong interest in using modern machine learning models to accelerate this simulation process while maintaining sufficient physical fidelity, which will then enable the production of larger simulated datasets needed for precision studies at the LHC as well as the High-Luminosity Large Hadron Collider (HL-LHC) scheduled to start operations in June 2030.

Stage	Description
Event generation	Physics event generators such as PYTHIA or HERWIG simulate the initial proton-proton collision. This stage records truth-level information and determines which particles are treated as stable enough to be propagated through the detector.
Detector simulation	The generated particles are passed through a simulated model of the ATLAS detector, where their interactions with detector material are modeled using tools such as GEANT4 or ATLFAST3. These interactions produce detector “hits,” which record quantities such as deposited energy, position, and time.
Digitization	The simulated hits are then converted into digital signals. At this stage, information from different simulated event components can be overlaid to approximate the conditions of real LHC data-taking.
Reconstruction	The digitized detector signals are processed with reconstruction algorithms similar to those used for real ATLAS data.

Table 1.1: Traditional ATLAS Monte Carlo simulation production chain.

1.1 Previous Works

Previous works have shown the effectiveness of machine learning in simulating particle events. In particular, CMS FlashSim [1], an end-to-end machine-learning-based simulation framework that maps generator-level events directly to CMS NANOAOD analysis-level objects, uses generative models, including normalizing-flow and flow-matching methods, to reproduce reconstructed physics objects and their correlations at much higher speed than conventional full simulation.

While CMS FlashSim demonstrates the promise of end-to-end machine-learning-based fast simulation, its success does not automatically establish that similar methods will work for ATLAS reconstructed jet data or for top quark mass reconstruction. This project therefore studies a more focused problem: whether normalizing flows can accurately model ATLAS jet-level observables and preserve the correlations needed to reproduce a physically meaningful derived quantity, the reconstructed top-quark mass. Rather than proposing a full replacement for the ATLAS simulation chain, this work serves as a targeted feasibility and validation study for flow-based fast simulation in an ATLAS jet-analysis setting.

1.2 Thesis Overview

This thesis studies the use of Normalizing Flows (NFs) as a generative model for ATLAS reconstructed jet data. The goal is to investigate to what extent a flow-based model can learn the distribution of jet-level observables and generate synthetic events that preserve physically meaningful correlations. As a validation task, the generated jets are used to reconstruct top quark masses, and the reconstructed top mass distribution is compared with that from the original simulated data.

The remainder of this thesis is organized as follows: Chapter 2 provides the relevant physics and machine learning background, including top quark pair production, top quark mass reconstruction, and normalizing flows. Chapter 3 introduces the dataset, preprocessing steps, model architecture, and training procedure. Then chapter 4 presents the evaluation of the generated jet distributions and the reconstructed top quark mass results. Finally, Chapter 5 discusses the findings, limitations, and possible directions for future work.

2. Background

2.1 Detector Coordinate System

In ATLAS, particles produced in the collisions are described using a coordinate system adapted to the geometry of the detector, as shown in Figure 2.1. The beam direction defines the z -axis, with the transverse plane perpendicular to the beam. Instead of using the polar angle directly, particles are often described by the pseudorapidity

$$\eta = -\ln \tan \left(\frac{\theta}{2} \right),$$

where θ is the angle measured from the beam axis. The azimuthal angle ϕ describes the direction of the particle in the transverse plane, and the transverse momentum p_T measures the component of momentum perpendicular to the beam direction.

2.2 Jets

A proton-proton collision can produce many final-state particles, but one of the most important reconstructed objects for this thesis is the jet. Jets are collimated sprays of particles produced by the hadronization of energetic quarks and gluons. Since quarks and gluons cannot be observed directly as free particles due to color confinement, a phenomenon in quantum chromodynamics, their presence is inferred from the clusters of particles they leave in the detector, that is, jets.

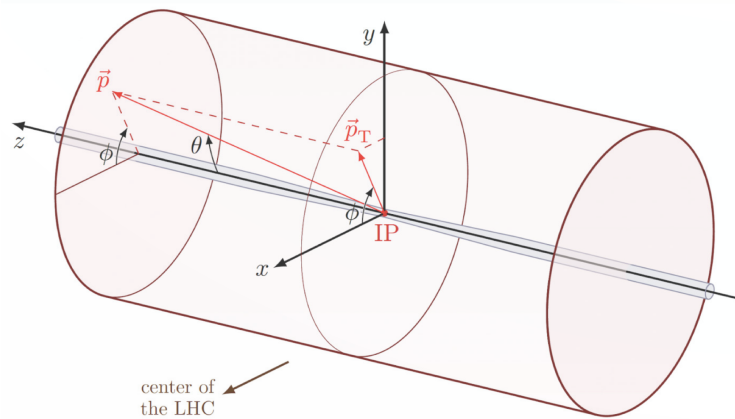


Figure 2.1: ATLAS detector coordinate system.

Jets in ATLAS are characterized by their coordinates (p_T, η, ϕ) and invariant masses. In addition, each jet is associated with four flavor-tagging scores, p_b, p_c, p_u , and p_τ , which represent the probabilities that the jet originated from a bottom quark, charm quark, light-flavor quark or gluon, or hadronic τ decay, respectively. These scores are generated by physics-informed machine learning algorithms, and they are important because identifying flavors of jets are essential for the reconstruction of events. For example, jets from bottom quarks are especially relevant for top quark mass reconstruction, since each top quark decays almost always into a W boson and a bottom quark, as we will discuss in more detail below. In this case, the flavor-tagging scores are used to identify the two most likely b -jets in each event, which are then combined with reconstructed W candidates to form top-quark candidates.

2.3 Top Quark Pair Production

The process we focus on in this thesis is top quark pair production, as outlined in the Feynman diagram in Figure 2.2. In particular, we look at the fully hadronic channel, where all final-state particles are jets. In the fully hadronic channel, both top quarks decay through $t \rightarrow Wb$ into a

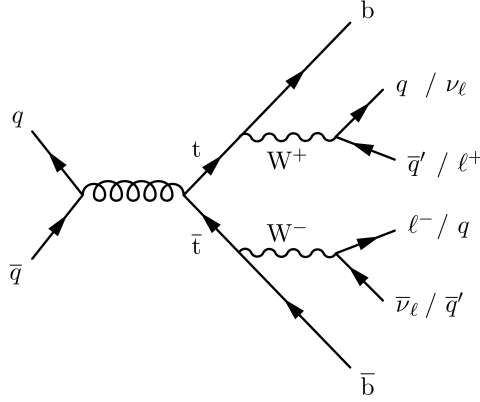


Figure 2.2: Top quark pair production.

W boson and a bottom quark. Then, both W bosons decay into quark-antiquark pairs. The decay chain can be written schematically as

$$t\bar{t} \rightarrow (W^+b)(W^-\bar{b}) \rightarrow (q\bar{q}'b)(q\bar{q}'\bar{b})$$

After hadronization, the six final-state quarks appear in the detector as jets: two b -jets from the top quark decays and four light-flavor jets from the two W bosons. This channel is common among $t\bar{t}$ events because hadronic decays of the W boson occur frequently, so a sizable fraction of top quark pair events produce an all-jet final state.

2.4 Top Quark Mass Reconstruction

To evaluate whether the generated jet events preserve physically meaningful structure, we use a simple reconstruction algorithm to reconstruct the masses of $t\bar{t}$ from final state jets $(q\bar{q}'b)(q\bar{q}'\bar{b})$. One thing to note is that although ideally the fully hadronic $t\bar{t}$ decay produces six primary quarks, additional reconstructed jets may arise from additional particles produced during the collision or from detector and reconstruction effects. Therefore, input to the algorithm often contains more than 6 final state jets.

The reconstruction algorithm first identifies the two jets with the largest p_b scores as the two b -jet candidates. The next step is to pair the remaining non- b jets into two W candidates. Since each W boson should have an invariant mass close to the known W boson mass, the algorithm considers all possible ways of selecting two disjoint pairs of jets and chooses the pairing whose invariant masses are closest to $m_W \approx 80.4$ GeV. For each jet pair, the invariant mass is computed by adding the four-momenta of the two jets and evaluating

$$m_{jj}^2 = E_{jj}^2 - |\vec{p}_{jj}|^2.$$

where jet energy and momentum are computed from p_T, η, ϕ , and mass (m) from the relations

$$p_x = p_T \cos \phi, \quad p_y = p_T \sin \phi, \quad p_z = p_T \sinh \eta, \quad E = \sqrt{p_x^2 + p_y^2 + p_z^2 + m^2}$$

After the two W candidate pairs are selected, each is combined with one of the two selected b -jet candidates to form two top-quark candidates. The reconstructed top mass is then computed from the invariant mass of each three-jet system,

$$m_{bjj}^2 = E_{bjj}^2 - |\vec{p}_{bjj}|^2.$$

This gives two reconstructed top-quark masses per event, corresponding to the two top candidates. This reconstruction procedure is not intended to be a full experimental top quark analysis, but rather a physics-motivated quality check for the generative model. Top quark mass reconstruction depends simultaneously on jet kinematics, flavor-tagging information, jet pairing, and multi-jet invariant masses. Therefore, agreement between the reconstructed top-mass distribution from generated events and that from the original simulated events provides a strong indication of whether the model has learned physically relevant structure in the data.

2.5 Normalizing Flows

Normalizing Flows [2] are a class of generative models. A normalizing flow is an invertible map from a simple distribution (e.g., a standard normal) to a more complex distribution (i.e., the target distribution from which we want to draw samples). Formally, define the latent space $\mathcal{Z} \subseteq \mathbb{R}^d$ with a latent variable $z \in \mathcal{Z}, z \sim \mathcal{N}(0, I_d)$ and the target space $\mathcal{X} \subseteq \mathbb{R}^d$ with a target variable $x \in \mathcal{X}, x \sim p_X(x)$, then a normalizing flow is an invertible map $\mathbf{f}_\theta : \mathcal{Z} \rightarrow \mathcal{X}$ parametrized by θ (which is learned by neural networks) such that $x = \mathbf{f}_\theta(z)$ and $z = \mathbf{f}_\theta^{-1}(x)$. Then the probability density in the target space can be evaluated using the Jacobian determinant

$$p_X(x) = p_Z(\mathbf{f}_\theta^{-1}(x)) \left| \frac{\partial \mathbf{f}_\theta^{-1}}{\partial x} \right|$$

To ensure model flexibility and training stability, the map $\mathbf{f}_\theta$ is defined to be the composition of a sequence of invertible transformations $\mathbf{f}_\theta = \mathbf{f}_{k, \theta_k} \circ \mathbf{f}_{k-1, \theta_{k-1}} \circ \dots \circ \mathbf{f}_{1, \theta_1}$, and it also makes sense mathematically because the Jacobian determinant of a composed function can be nicely written as a product of Jacobian determinants by the chain rule. Therefore, fast density evaluation and training can be achieved by efficient computation of the Jacobian determinants, which is why in practice we choose functions $\mathbf{f}_{i, \theta_i}$ such that $\partial \mathbf{f}_{i, \theta_i}^{-1} / \partial x$ is triangular so that the Jacobian determinant is simply the product of its diagonal elements.

In this thesis, we use a type of normalizing flows called Rational Quadratic Spline Flows [3]. Let

$$\mathbf{f}_\theta = (f_{\theta^{(1)}}, f_{\theta^{(2)}}, \dots, f_{\theta^{(d)}}) \quad \text{where} \quad y_j = f_{\theta^{(j)}}(x_j)$$

then each $f_{\theta^{(j)}}$ is the composition of a sequence of splines with K bins, each bin being represented by a monotonically increasing rational-quadratic function. The boundaries of these functions are

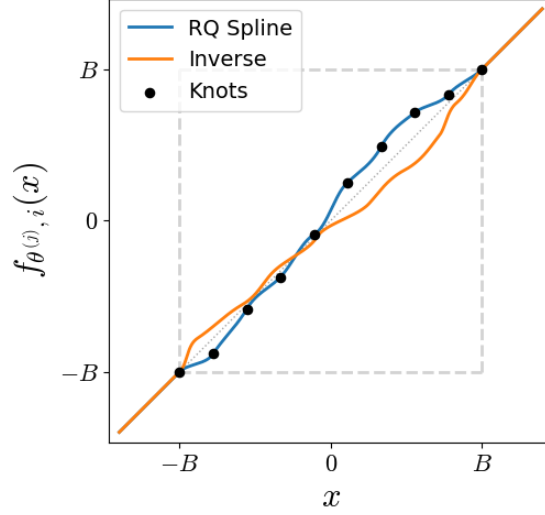


Figure 2.3: A monotonically increasing rational-quadratic spline parametrized by 8 knots in $[-B, B] \times [-B, B]$ and 8 derivatives at each knot.

determined by $K + 1$ knots $\{(x^{(k)}, y^{(k)})\}_{k=0}^K$ in $[-B, B] \times [-B, B]$, and there is a derivative associated with each internal knot (see Figure 2.3). Here, K and B are both hyper-parameters.

In addition to choosing a class of invertible functions, our model uses an autoregressive structure to make the high-dimensional map tractable. In an autoregressive flow, each parameter $\theta^{(l)}$ is produced by a neural network using only $x_1, \dots, x_{l-1}$ as inputs, and this restriction naturally ensures that the Jacobian matrix is triangular. It is worth to note that in our case, the autoregressive neural network does not directly output the transformed value y_j . Instead, it outputs the spline parameters which define the monotone transformation applied to x_i .

Finally, we use a conditional normalizing flow. Rather than modeling only the marginal distribution $p_X(x)$ of reconstructed jet features, the model learns a conditional distribution $p_X(x | c)$, where c denotes context variables from the truth-level jets. In this way, the generated reconstructed jets are encouraged to remain consistent with the underlying simulated event.

3. Methodology

3.1 Dataset Overview

The dataset used for this thesis is ATLAS $t\bar{t}$ simulation for machine-learning-based jet flavor tagging (JetSet) [4], released through the CERN Open Data Portal. The dataset consists of simulated proton-proton collision events producing top-quark pairs at a center-of-mass energy of $\sqrt{s} = 13.6$ TeV. It was originally designed for developing and evaluating machine learning methods for jet flavor tagging. The full release contains approximately 50 million simulated events and is stored in HDF5 format, with structured event-level, jet-level, track-level, and truth-level information. For this thesis, we use only a subset of the available jet-level and truth-level variables relevant to generative modeling and top-quark mass reconstruction.

In the dataset, each event contains a variable number of reconstructed jets. For each reconstructed jet, we use its transverse momentum p_T , pseudorapidity η , azimuthal angle ϕ , and mass m , which provide the information needed to reconstruct four-momenta and compute invariant masses, and the flavor-tagging scores p_b , p_c , p_u , and p_τ . The dataset also provides truth-level variables p_T^{truth} , η^{truth} , ϕ^{truth} , and m^{truth} . These variables will be used as the context in our conditional flow model. Finally, the variable `eventNumber` is used to group jets belonging to the same collision event. This grouping is essential because the normalizing flow model and the top-quark reconstruction algorithm operate at event level rather than treating jets as completely independent objects.

In summary, the variables used are

$$\text{target} = (p_T, \eta, \phi, m, p_b, p_c, p_u, p_\tau), \quad \text{context} = (p_T^{\text{truth}}, \eta^{\text{truth}}, \phi^{\text{truth}}, m^{\text{truth}})$$

grouped by `eventNumber`.

3.2 Data Preprocessing

Before training the model, there are some preprocessing steps that can help stabilize training and improve model performance. The transverse momentum and mass in the dataset are originally given in MeV. We convert p_T , m , p_T^{truth} , and m^{truth} to GeV by dividing by 1000. Events and jets with missing values in the selected columns are removed. We then apply basic kinematic cuts to select jets satisfying $p_T > 30$ GeV, $|\eta| < 2.5$. These cuts restrict the dataset to energetic central jets, which are more relevant for the top-quark reconstruction procedure. After this selection, we keep only events containing at least six reconstructed jets, since the fully hadronic $t\bar{t}$ final state is expected to contain six primary jets: two b -jets and four jets from W boson decays.

Several transformations are applied to make the variables more suitable for training with a normalizing flow. First, we use $\log(p_T)$ and $\log(m)$, which compresses large values and spreads out smaller values, making their distributions less sharply peaked and easier for the normalizing flow to model. Next, for the angular variables, we transform η using $\tan(\eta/2)$ and represent the periodic azimuthal angle ϕ by $\cos(\phi), \sin(\phi)$. This avoids the discontinuity at $\phi = \pm\pi$ and allows the model to learn the angular dependence more smoothly.

The four flavor-tagging scores are also transformed before training. Rather than modeling all four

probabilities directly, we use log-ratio variables relative to the light-flavor score p_u :

$$z_b = \log \left(\frac{p_b}{p_u} \right), \quad z_c = \log \left(\frac{p_c}{p_u} \right), \quad z_\tau = \log \left(\frac{p_\tau}{p_u} \right).$$

This reduces the four constrained probability values to three unconstrained real-valued variables. The original flavour-tagging score columns are then removed from the training features.

After feature transformations, jets are sorted within each event by decreasing reconstructed p_T . This imposes a consistent ordering on the jets, allowing each event to be represented as a fixed-length vector rather than as an unordered set. The events are then randomly split into training and testing sets using an 80/20 split at event level.

Finally, for each event, the selected reconstructed jet features are concatenated into a single fixed-length vector corresponding to six jets, where the six jets are chosen in the same way as in the top mass reconstruction algorithm. This produces the target 48-dimensional input x for the normalizing flow. The truth-jet variables are stored separately as the conditioning information c . Thus, the model is trained to learn the conditional distribution $p(x \mid c)$, where x represents the reconstructed event-level jet features and c represents the corresponding truth-level jet information.

3.3 Model Setup and Training

We mainly used the PyTorch package NORMFLOWS [5] to build our model.

Before training, the reconstructed target vectors are standardized feature-wise using the mean and standard deviation computed from the training set. This places the input features on comparable numerical scales, which improves optimization stability. The conditional variables are provided to

the model as context vectors. During sampling, generated standardized samples are transformed back to the original preprocessed feature scale.

The base distribution of the flow is a 48-dimensional diagonal multivariate Gaussian. The flow consists of autoregressive rational-quadratic spline layers, with permutation layers in between to mix the feature ordering across dimensions. This helps ensure that the autoregressive structure does not always model the variables in the same fixed order.

The model is trained by minimizing the forward KL divergence. That is, the model is trained to maximize the likelihood of the corresponding reconstructed jet features under the learned conditional distribution. Training is performed using the Adam optimizer with batch size 1024.

The final model used consists of 12 flow layers. Each autoregressive network has 3 hidden layers with 128 hidden units per layer. Each spline transformation uses 16 bins and a tail bound $B = 3$. The model is trained with learning rate 10^{-6} and weight decay 5×10^{-5} . After training, new reconstructed event samples are generated by drawing from the base Gaussian distribution and applying the inverse conditional flow using the corresponding truth-level event vector as context.

4. Results

4.1 Kolmogorov-Smirnov Statistic

To quantify the similarity between the test and generated distributions, we use the Kolmogorov-Smirnov (KS) statistic. Given the empirical cumulative distribution functions $F_{\text{test}}(x)$, $F_{\text{gen}}(x)$ for the distributions, the KS statistic is defined as

$$D_{\text{KS}} = \sup_x |F_{\text{test}}(x) - F_{\text{gen}}(x)|.$$

where a smaller KS statistic indicates better agreement between the two distributions.

4.2 Results and Discussions

To evaluate the quality of the generated events, we compare the generated reconstructed jets with the test dataset at several levels. Since the goal is not only to reproduce each individual feature but also to preserve correlations between jets and physically meaningful derived quantities, we use three main types of evaluation: individual feature distributions, feature correlation matrices, and reconstructed top-quark mass distributions.

First, we compare the one-dimensional distributions of each generated feature against the corresponding distribution in the test set. These include the reconstructed jet kinematic variables p_T ,

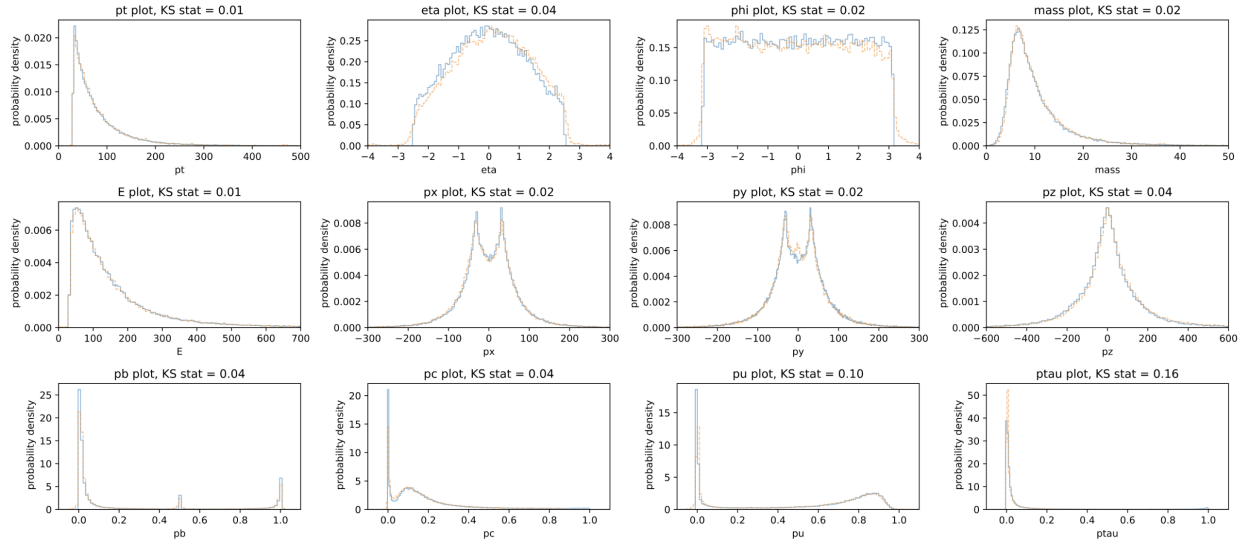


Figure 4.1: Generated vs testing individual feature distributions.

η , ϕ , and m , as well as the flavor-tagging scores p_b , p_c , p_u , and p_τ after transforming generated samples back to the physical feature space. The results are shown in Figure 4.1. As we can see, the generated samples agree well with the test data for the kinematic variables, including their sharp peaks and long tails, with small KS statistics below 0.05. The angular variables are also modeled reasonably well: the generated η distribution captures the central peak and detector acceptance cutoff, while the ϕ distribution remains approximately uniform. The main discrepancies occur in the flavor-tagging scores, especially p_u and p_τ , whose KS statistics are larger. These variables are highly peaked and nearly discrete, which are more difficult for a continuous normalizing flow to reproduce exactly. Overall, the model captures the marginal kinematic structure of the data well, while the tagging-score distributions are more challenging to reproduce.

Agreement in the feature distributions indicates that the normalizing flow has learned the marginal distribution of each variable. However, matching individual feature distributions alone is not sufficient, because a model can reproduce each feature separately while still failing to capture correlations between them. To study correlations, we compute Pearson correlation matrices for both the

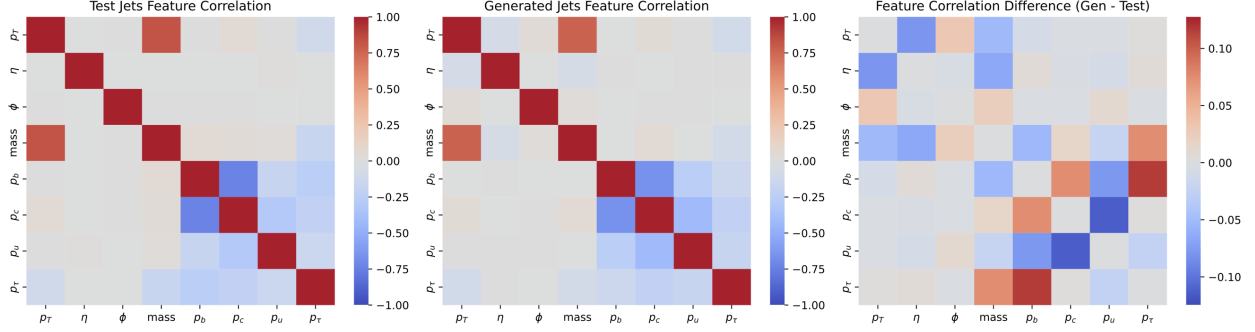


Figure 4.2: Test, generated, and difference correlation matrices.

test events and the generated events. For a set of event-level features $\{X_1, \dots, X_d\}$, the Pearson correlation coefficient between features X_i and X_j is

$$\rho_{ij} = \frac{\text{Cov}(X_i, X_j)}{\sigma_i \sigma_j}.$$

We visualized the correlation matrix of the test data, the correlation matrix of the generated data, and their difference, $\Delta\rho_{ij} = \rho_{ij}^{\text{gen}} - \rho_{ij}^{\text{test}}$, as shown in Figure 4.2. The feature correlation matrices show that the generated samples reproduce the main correlation structure of the test data reasonably well. In particular, the strong positive correlation between p_T and mass is preserved, as are the negative correlations among some of the flavor-tagging scores, especially between p_b and p_c . The difference matrix shows that most correlation differences are relatively small, indicating that the model captures the overall multivariate structure of the data. The largest deviations appear in correlations involving the flavor-tagging scores. This is consistent with the individual feature distributions, where the tagging scores were also more difficult to model accurately. Overall, the correlation comparison suggests that the model learns the dominant kinematic relationships well, while some finer structure in the flavor-tagging variables remains imperfect.

Finally, we evaluate the generated samples using the reconstructed top quark mass distribution. For both the test data and the generated data, we apply the same top-quark reconstruction algorithm

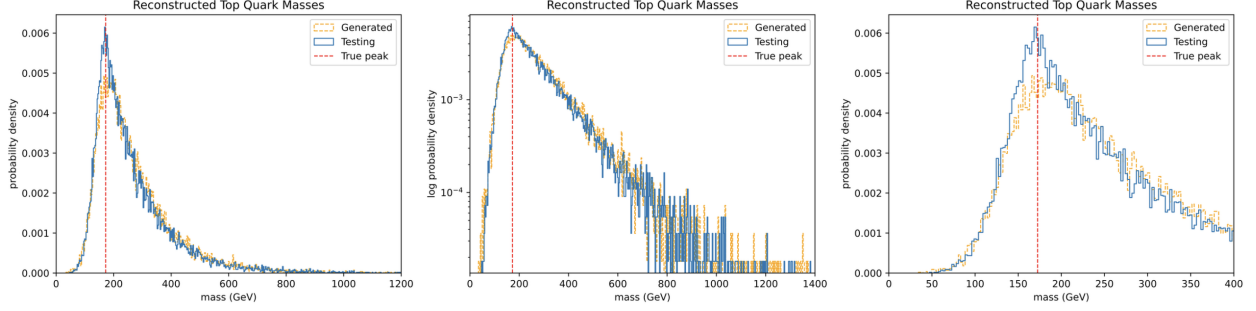


Figure 4.3: Test vs generated reconstructed top quark mass distributions. Normal scale (left); log scale (middle); and zoomed in view of normal scale (right).

described above. The distributions of reconstructed top quark masses for the test and generated samples are shown in Figure 4.3. The reconstructed top-quark mass distributions show good overall agreement between the generated and test samples. The generated distribution reproduces the main peak near the expected top-quark mass and follows the broad asymmetric shape of the test distribution, including the long tail. The log-scale plot confirms that the agreement remains reasonable over several orders of magnitude in probability density, although small deviations are visible in the far tail where statistics are lower.

Across five independent trials, the reconstructed top-quark mass distributions achieved an average KS statistic of 0.058 ± 0.021 . This indicates reasonably good agreement between the generated and test distributions, since the maximum difference between their empirical cumulative distribution functions is typically only about 6%. The nonzero standard deviation shows that the reconstruction quality varies somewhat between trials, likely due to stochastic training and sampling effects. Overall, the result suggests that the model can reproduce the main structure of the top-mass distribution, including the peak region and overall shape, although some residual discrepancies remain.

5. Conclusion

In this thesis, we investigated the use of conditional normalizing flows as a fast, data-driven approach for modeling reconstructed jets at the ATLAS experiment. The model was trained to generate reconstructed jet features from truth-level jet information, with the broader motivation of exploring machine-learning-based alternatives to computationally expensive simulation workflows.

The results demonstrated that normalizing flows are a promising class of generative models for modeling reconstructed jet distributions. In particular, the model was able to reproduce most one dimensional features with small KS-statistics. However, discrepancies in correlation matrices and the reconstructed top quark mass distribution indicate fast simulation that preserves detailed correlations, flavor-tagging behavior, and event-level physics information remains challenging.

There are several natural directions for future work. One important extension could be to improve the event representation. In this work, events were modeled by concatenating jets in a fixed order, which allows the model to learn event-level information but also imposes a particular structure on the input. Future studies could explore permutation-invariant architectures that more naturally handle a variable number of jets, such as set-based models, graph neural networks, or transformer-based event representations. These approaches could allow the model to learn relationships among jets without depending on a manually chosen ordering.

Another possible improvement is to also train the model on track-level and vertex-level features, and feed them into a working flavor-tagging model to obtain the tagging scores, instead of forcing the model to directly learn their nearly-discrete distributions. Future work could also expand the validation strategy, such as testing the model on other reconstructed physics quantities.

References

- [1] Andrea Rizzi, Francesco Vaselli, Filippo Cattafesta, and Patrick Asenov. CMS flashsim: end-to-end simulation with machine learning. 2024.
- [2] Ivan Kobyzev, Simon J.D. Prince, and Marcus A. Brubaker. Normalizing flows: An introduction and review of current methods. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(11):3964–3979, November 2021.
- [3] Conor Durkan, Artur Bekasov, Iain Murray, and George Papamakarios. Neural spline flows, 2019.
- [4] ATLAS Collaboration. Transforming jet flavour tagging at ATLAS. *Nature Communications*, 17(1), January 2026.
- [5] Vincent Stimper, David Liu, Andrew Campbell, Vincent Berenz, Lukas Ryll, Bernhard Schölkopf, and José Miguel Hernández-Lobato. normflows: A pytorch package for normalizing flows. *Journal of Open Source Software*, 8(86):5361, 2023.